Sri Sivasubramaniya Nadar College of Engineering

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajiv Gandhi Salai (OMR), Kalavakkam – 603110

Project Work
(Phase 1 & Phase 2)
Academic Year: 2026-2027
Batch: 2023-2027

Project Team Members

Thirumurugan RA (3122235001149)

Meghna Manimaran (3122235001079)

Shrika SP Thota (3122235001125)

Supervisor

Dr. N. Sujaudeen

Assistant Professor

B.E
Computer Science and Engineering

Regulations 2021



COURSE CODE	COURSE TITLE	L	T	P	E	C
UCS2717	PROJECT WORK PHASE I	0	0	6	0	3

OBJECTIVES

- To develop skills to formulate a complex engineering problem with an awareness of issues pertaining to society, health, safety, legal, environment, culture and examine the impact of the proposed solution on these issues.
- To carry out the survey on related literature and find out the research gap
- To explore use of new tools, algorithms and techniques required to carry out the projects.
- To enable the students to gain experience in organization and implementation of a project and acquire the necessary confidence to carry out the extensions of the project in the phase 2.
- To follow the guidelines and prepare technical report of the project.
- To develop team management skills and presentation skills

COURSE OUTCOMES

On successful completion of this course, the student will be able to:

1. Formulate complex engineering problems and choose suitable technologies and tools required to solve them effectively (K4)
2. Analyse the related literature on the proposed problem and its feasibility (K4)
3. Design high level solutions by applying software engineering principles (K6)
4. Divide the work appropriately among the team members to manage time and complexity.
5. Document and present the solution.
6. Analyse the issues pertaining to society, health, safety, legal, environment, culture applicable to the project and examine the impact of the proposed solution on these issues.
7. Apply best practices and follow ethics in developing the solutions.

Evaluation: 60:40

R1		R2		R3		End Semester Examination			
						Project Report	Viva Voce		
Committee	Guide	Committee	Guide	Committee	Guide		Internal	External	Viva
10	10	10	10	10	10	10	10	10	10

Evaluation criteria for each review

Review 1	Marks	Review 2	Marks	Review 3	Marks
1.Problem Formulation 2. Justification of the formulation 3. Literature Review and feasibility study 4. Apply knowledge to solve problem 5. Design Formulation	70%	1.Design choice 2.Design Methodology 3.Design Solution 4.Develop and Conduct Experiment 5. Design of system evaluation	70%	1.Design Solution 2.Design Improvement 3. Project demonstration 4.Project Innovation	70%
1.Project management 2.Oral Presentation 3.Presentation: Knowledge and Theory 4.Presentation: Interaction 5.Team Project Execution and Effort 6.Written Report 7.Plagiarism in the Report	30%	1.Team Motivation. 2.Describe the issues pertaining to society, health, safety, legal, environment, culture applicable to the project 3. Written report and presentation: Interaction 4.Plagiarism in the Report	30%	1. Demonstrate the knowledge of, and need for sustainable development 2. Team Project Execution and Effort 3. Project report 4. Plagiarism in the Report	30%

Detailed rubrics for three reviews are given below:
Rubrics for Review 1

S. No	Categories	Score 3	Score 2	Score 1	Score 0
1	Problem Formulation (CO1)	The complex engineering problem formulation and concept is very clear.	The complex engineering problem formulation and concept is moderately. clear.	The complex engineering problem formulation and concept have basic level clarity.	The problem formulation and concept are vague and unclear.
2	Justification of the formulation (CO1)	Complexity of the problem, volume of the work, scope and motivation of the work are clearly justified	Complexity of the problem, volume of the work are justified	Complexity of the problem is justified.	Justification of the problem formulation is not done.
3	Literature Review and feasibility (CO2)	Outstanding synthesis of information from peer-reviewed research articles is done. Gaps arising from the studies are clearly identified. Feasibility of implementing the proposed problem is analysed in depth.	Peer-reviewed research articles are summarized effectively. Gaps arising from the studies are identified. Feasibility of implementing the proposed problem is analysed at the satisfactory level	Peer-reviewed research articles are summarized, but not all are directly relevant to the specific topic . Gaps arising from the studies are not identified. Feasibility of implementing the proposed problem is not done.	Peer- reviewed research articles are not listed. Summary of information from these articles is absent. Feasibility of implementing the proposed problem is not done.
4	Apply knowledge of mathematics, basic science and engineering (CO1)	The knowledge of mathematics, basic science and engineering applied to solve the problem is sound.	The knowledge of mathematics, basic science and engineering applied to solve the problem is reasonable.	The knowledge of mathematics, basic science and engineering applied to solve the problem is having some minor errors.	The knowledge of mathematics, basic science and engineering is not applied to solve the problem.
5	Design Formulation (CO3)	The design formulation considers relevant constraints and specifications.	The design formulation considers the majority of relevant constraints and most specifications.	The design formulation considers only a smaller number of relevant constraints and specifications.	The design formulation is not correct as constraints and specifications are not understood properly.
6	Project management (CO3)	Has understood the project scope well and presented an effective plan.	Has understood the project scope and presented a not very effective but achievable plan.	Has partially understood the project scope and presented a sketchy plan.	Not much understanding of project scope and no plan was presented.
7	Oral Presentation (CO5) (For individual team member)	The oral presentation is well structured, clear, fluent and confident.	The oral presentation is well structured and clear.	The oral presentation is clear	The oral presentation is poor.
8	Presentation: Knowledge and Theory (CO5)	The student shows excellent grasp of overall and project specific fundamentals as well as advanced engineering knowledge and theory.	The student shows a good grasp of overall and project specific fundamentals as well as advanced engineering knowledge and theory.	The student shows basic level grasp of overall and project specific fundamentals in engineering knowledge and theory. .	The student has a poor grasp of overall and project- specific fundamentals
9	Presentation: Interaction (CO5) (Individual team member)	Students understand the questions and answer confidently.	Students understand the questions and answer them satisfactorily.	Students understand the questions and answer them wrongly.	Students did not understand the question.
10	Team Project Execution and Effort (CO4)	The students meet their guide on a regular basis and follow timelines as planned.	The students meet their guide on a regular basis, but do not meet the timelines as planned.	The students meet their guide only sporadically	The students did not meet the guide after the previous review.
11	Written Report and presentation slides (CO5)	Grammatical and spelling corrections are carried out. The qualities of the figure, reference format are correct. Citation of tables, figures, and references are given correctly.	Grammatical and spelling corrections are carried out. The qualities of the figure, reference format are correct.	Grammatical and spelling corrections are carried out.	The report and presentation slides have lot of errors
12	Plagiarism in the Report (CO5)	There is no plagiarism and substantial improvement from the previous report.	There is no plagiarism but limited improvement from the previous report.	There is reasonable plagiarism. No improvement from the previous report.	Most of the report content is plagiarized

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Department of Computer Science and Engineering

Project Evaluation Sheet – REVIEW ONE

Project Discussion

Note: Project Hours / Week = 15 Hours

Week	Dates	Discussed	Week Attendance	Signature (Guide)
1				
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10				

Total Project Hours discussed:

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Project Evaluation Sheet – REVIEW ONE

Rubrics (Score 0 – 3)

1. Problem Formulation
2. Justification of the problem
3. Literature review and feasibility
4. Apply knowledge of mathematics, basic science and engineering
5. Design Formulation
6. Project management
7. Oral Presentation
8. Presentation: Knowledge and Theory
9. Presentation: Interaction
10. Team Project Execution and Effort
11. Written Report and presentation slides
12. Plagiarism in the Report

Reg. No	1	2	3	4	5	6	7	8	9	10	11	12

Review comments: (to be filled by students)

Signature of the students **Signature of the supervisor**

Rubrics – Project Review Two
Rubrics for Review 2

S. No	Categories	Score 3	Score 2	Score 1	Score 0
1	Design Choice (CO3)	The most appropriate choice is made after a thorough analysis.	A reasonable choice is made after a good analysis.	A choice is made after basic analysis.	The misunderstanding may lead to irrelevant choices.
2	Design Methodology (CO3)	The methodology to implement the design is sound.	The methodology to implement the design is reasonable.	The methodology to implement the design is mostly correct but has minor errors.	The methodology to implement is mostly wrong.
3	Develop and Conduct Experiment (CO5)	The implementation and the experiments are scientifically sound.	The implementation and the experiments are scientifically sensible.	The implementation and the experiments are having some minor errors.	The implementation and the experiments are wrong.
4	Design of system evaluation (CO5)	The methodology for system evaluation is appropriate.	The methodology for system evaluation is reasonable.	The methodology for system evaluation is mostly correct but has minor errors.	The methodology for system evaluation is not identified.
5	Team Project Execution and Effort (CO4)	The students meet their guide on a regular basis and follow timelines as planned.	The students meet their guide on a regular basis, but do not meet the timelines as planned.	The students meet their guide only sporadically	The students did not meet the guide after the previous review.
6	Describe the issues pertaining to society, health, safety, legal, environment, Culture applicable to the project (CO6)	have a very good understanding of the issues and can relate to the project	have a relatively good understanding of the issues and can partly relate to the project	Can partly understand these issues but cannot relate to their project	there is no understanding of these issues
7	Written Report and presentation interaction (CO5)	All technical aspects of the project are covered and documented as per the format. All the questions from the evaluation panel are answered confidently.	All technical aspects of the project are covered. Most of the questions asked by the evaluation panel are answered confidently.	Technical aspects of the project are partially covered. The questions are not answered satisfactorily.	The report and presentation slides have lot of errors. The questions are not at all answered or answered wrongly.
8	Plagiarism in the Report (CO5,CO7)	There is no plagiarism and substantial improvement from the previous report.	There is no plagiarism but limited improvement from the previous report.	There is reasonable plagiarism. No improvement from the previous report.	Most of the report content is plagiarized

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Project Evaluation Sheet – REVIEW TWO

Project Discussion

Note: Project Hours / Week = 15 Hours

Week	Dates	Discussed	Week Attendance	Signature (Guide)
1				
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Total Project Hours discussed:

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Project Evaluation Sheet – REVIEW TWO

Rubrics (Score 0 – 3)

1. Design Choice
2. Design Methodology
3. Develop and Conduct Experiment
4. Design of system evaluation
5. Team Project Execution and Effort
6. Describe the issues pertaining to society, health, safety, legal, environment,
7. Culture applicable to the project
8. Written Report and presentation interaction
9. Plagiarism in the Report

Reg. No	1	2	3	4	5	6	7	8	9

Review comments: (to be filled by students)

Signature of the students **Signature of the supervisor**

Rubrics - Project review 3
Rubrics for Review 3

S. No	Categories	Score 3	Score 2	Score 1	Score 0
1	Design Solution (CO3)	The implemented solution is sound.	The implemented solution is very good.	The implemented solution is good.	The implementation is unreasonable and flawed.
2	Design Improvement (CO3)	The design iteratively modified for improvement.	One iteration is done on the design for improvement.	An iteration is proposed for improvement of the design but not implemented.	The design unable to function and no attempts to correct.
3	Project Demonstration (CO5)	·All defined objectives are achieved. ·Each module working well and properly demonstrated. ·All modules of project are well integrated and performance analysis carried out	·All defined objectives are achieved. ·Each module working well and properly demonstrated ·Integration of all modules not done, and performance analysis is not very satisfactory	·All defined objectives are achieved. ·Modules are working well in isolation and properly demonstrated. ·Modules of project are not properly integrated	· Only some of the defined objectives are achieved. · Modules are not in proper working form that further leads to failure of integrated system
4	Project Innovation (CO5)	The results have many innovative elements.	The results have some innovative elements.	The results might lead to minor confusion in personal practice.	The results lead to major personal confusion in practical understanding.
5	Demonstrate the knowledge of, and need for sustainable development (PO6)	The knowledge of sustainability (i.e reusability, developed system is sustainable for technology change etc) related to their project work is sound	The knowledge of sustainability (i.e reusability, developed system is sustainable for technology change etc) related to their project work is reasonable.	The knowledge of sustainability (i.e reusability, developed system is sustainable for technology change etc) related to their project work is poor	The knowledge of sustainability (i.e reusability, developed system is sustainable for technology change etc) related to their project work is not there
6	Team Project Execution and Effort (CO4)	The students meet their guide on a regular basis and follow timelines as planned.	The students meet their guide on a regular basis, but do not meet the timelines as planned.	The students meet their guide only sporadically	The students did not meet the guide after the previous review
7	Project Report (CO5)	·Project report is according to the specified format ·References and citations are appropriate and well mentioned	·Project report is according to the specified format. ·References and citations are appropriate but not mentioned well	·Project report is according to the specified format but some mistakes. ·In-sufficient references and citations	·Project report not prepared according to the specified format. ·References and citations are not appropriate
8	Plagiarism in the Report (CO5, CO7)	There is no plagiarism and substantial improvement from the previous report.	There is no plagiarism but limited improvement from the previous report.	There is reasonable plagiarism. No improvement from the previous report.	Most of the report content is plagiarized

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Project Evaluation Sheet – REVIEW THREE

Project Discussion

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Project Evaluation Sheet – REVIEW THREE

Rubrics (Score 0 – 3)

1. Design Solution
2. Design Improvement
3. Project Demonstration
4. Project Innovation
5. Demonstrate the knowledge of, and need for sustainable development
6. Team Project Execution and Effort
7. Project Report
8. Plagiarism in the Report

Reg. No	1	2	3	4	5	6	7	8

Review comments: (to be filled by students)

Signature of the students **Signature of the supervisor**

COURSE CODE	COURSE TITLE	L	T	P	E	C
UCS2818	PROJECT WORK PHASE II	0	0	16	0	8

OBJECTIVES

- To make use of the knowledge gained by the student at various stages of the degree course.
- To develop the ability to solve a specific problem right from its identification and literature review till the successful solution of the same.
- To train the students in preparing project reports and to face reviews and viva voce examination.

COURSE OUTCOMES

On successful completion of this course, the student will be able to:

1. Use suitable technologies and tools to solve complex engineering problems effectively. (K4)
2. Analyze the related literature on the proposed problem and its feasibility (K4)
3. Construct suitable solutions by applying software engineering principles (K6)
4. Divide the work appropriately among the team members to manage time and complexity.
5. Evaluate the performance of the solution using appropriate metrics (K5)
6. Analyze the impact of the proposed solution on the issues pertaining to society, health, safety, legal, environment, culture and apply ethical principles.
7. Apply ethical principles in developing the solutions and present the solution with good documentation.

Evaluation criteria for each review

Review 1	Marks	Review 2	Marks	Review 3	Marks
Problem definition for Phase 2 - 15 Literature review for phase 2 - 15 Architecture diagram - 10 Detailed design - 15 Methodology implementation - 15	70%	Detailed design - 15 Module design by each candidate - 15 Methodology implementation - 15 Module Results - 15 Output analysis and metrics - 10	70%	Completeness and functionality -15 Evaluation and testing - 15 Performance analysis -15 Project demonstration - 15 Novelty - 10	70%
Project management Oral Presentation Written Report and presentation slides	30%	Describe the issues pertaining to society, health, safety, legal, environment, culture applicable to the project Oral Presentation Written report and presentation interaction	30%	Oral Presentation Project report Manuscript preparation	30%

Rubrics for Review 1

S. No	Categories	Score 3	Score 2	Score 1	Score 0
1	Project management (CO4)	Has understood the project scope well and presented an effective plan.	Has understood the project scope and presented a not very effective but achievable plan.	Has partially understood the project scope and presented a sketchy plan.	Not much understanding of project scope and no plan was presented.
2	Oral Presentation (For individual team member) (CO7)	The oral presentation is well structured, clear, fluent and confident.	The oral presentation is well structured and clear.	The oral presentation is clear	The oral presentation is poor.
3	Written Report and presentation slides (CO7)	Grammatical and spelling corrections are carried out. The qualities of the figure, reference format is correct. Citation of tables, figures, and references are given correctly.	Grammatical and spelling corrections are carried out. The qualities of the figure, reference format is correct.	Grammatical and spelling corrections are carried out.	The report and presentation slides have lot of errors

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Project Evaluation Sheet – REVIEW ONE

Project Discussion

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Project Evaluation Sheet – REVIEW ONE

Reg. No	Attendance (6)	Project Planning (3)	Project Execution (3)	Technical Contribution (3)

Review comments: (to be filled by students)

Signature of the students **Signature of the supervisor**

Rubrics – Project Review Two

S. No	Categories	Score 3	Score 2	Score 1	Score 0
1	Analyse the impact of the proposed solution on the issues pertaining to society, health, safety, legal, environment, culture and apply ethical principles (CO6)	have a very good understanding of the issues and can relate to the project	have a relatively good understanding of the issues and can partly relate to the project	Can partly understand these issues but cannot relate to their project	there is no understanding of these issues
2	Oral Presentation (For individual team member) (CO7)	The oral presentation is well structured, clear, fluent and confident.	The oral presentation is well structured and clear.	The oral presentation is clear	The oral presentation is poor.
3	Written Report and presentation interaction (CO7)	All technical aspects of the project are covered and documented as per the format. All the questions from the evaluation panel are answered confidently.	All technical aspects of the project are covered. Most of the questions asked by the evaluation panel are answered confidently.	Technical aspects of the project are partially covered. The questions are not answered satisfactorily	The report and presentation slides have lot of errors. The questions are not at all answered or answered wrongly.

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Project Evaluation Sheet – REVIEW TWO

Project Discussion

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Total Project Hours discussed:

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Project Evaluation Sheet – REVIEW TWO

Reg. No	Attendance (6)	Project Planning (3)	Project Execution (3)	Technical Contribution (3)

Review comments: (to be filled by students)

Signature of the students **Signature of the supervisor**

Rubrics – Project Review Three

S. No	Categories	Score 3	Score 2	Score 1	Score 0
1	Oral Presentation (For individual team member) (CO5)	The oral presentation is well structured, clear, fluent and confident.	The oral presentation is well structured and clear.	The oral presentation is clear	The oral presentation is poor.
2	Project Report (CO5)	·Project report is according to the specified format ·References and citations are appropriate and well mentioned	·Project report is according to the specified format ·References and citations are appropriate but not mentioned well	·Project report is according to the specified format but some mistakes ·In-sufficient references and citations	·Project report not prepared according to the specified format ·References and citations are not appropriate
3	Manuscript or paper (CO5)	Manuscript is submitted for conference or journal.	Manuscript is prepared for conference or journal. Review needs to done by the supervisor	Manuscript is not completed	Manuscript is not Prepared

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Project Evaluation Sheet – REVIEW THREE

Project Discussion

Note: Project Hours / Week = Hours

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Total Project Hours discussed:

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Project Evaluation Sheet – REVIEW THREE

Reg. No	Attendance (6)	Project Planning (3)	Project Execution (3)	Technical Contribution (3)

Review comments: (to be filled by students)

Signature of the students **Signature of the supervisor**